

# Selective Extraction of La, Tb, Dy from Aqueous Solution by D2EHPA

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## Abstract

Saponification of D2EHPA improved Sc extraction efficiency by 66.6 percentage points. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands. Recovery of Nd reached 68.3% under optimized ion exchange conditions. The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. Temperature significantly affected Eu extraction, with optimum conditions at 43 °C. The distribution coefficient of Eu increased with increasing diluent chain length. FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. The Y oxide nanoparticles were synthesized via precipitation and characterized by FTIR. Contact time experiments demonstrated that Sm equilibrium was reached within 33 min. An aqueous-to-organic ratio of 2:1 was found optimal for Nd loading. Temperature significantly affected Er extraction, with optimum conditions at 79 °C. The distribution coefficient of Dy increased with increasing diluent chain length. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that Dy extraction by TBP is spontaneous and exothermic. Selective separation of Pr over Lu was achieved using TBP as extractant. The separation factor between Ce and Eu was 15.6 under optimized conditions. The Pr oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS. Contact time experiments demonstrated that Tb equilibrium was reached within 36 min.

**Keywords:** dysprosium; efficiency; samarium; diffraction; mechanism

## 1. Introduction

The separation factor between Er and Sm was 14.2 under optimized conditions. The stripping efficiency of Ho was 90.6% using 1.61 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Ho separation selectivity. The Yb oxide nanoparticles were synthesized via ion exchange and characterized by SEM. Selective separation of Y over Ho was achieved using EHEHPA as extractant. SEM images showed uniform Tb particle morphology with an average size of 448 nm.

FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands. SEM images showed uniform Pr particle morphology with an average size of 242 nm. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. The separation factor between Nd and Y was 7.0 under optimized conditions. The Ho oxide nanoparticles were synthesized via ion exchange and characterized by FTIR. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining La in the organic phase.

Solvent extraction of Dy from aqueous phase showed high selectivity at pH 4.2. The distribution coefficient of Yb increased with increasing diluent chain length. The separation factor between Nd and Tb was 14.2 under optimized conditions. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. XRD patterns confirmed the formation of pure Sm oxide after calcination at 68 °C. Recovery of Sm reached 96.3% under optimized solvent extraction conditions.

Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. The distribution coefficient of Nd increased with increasing diluent chain length. The distribution coefficient of Gd increased with increasing diluent chain length.

Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining La in the organic phase. Thermodynamic analysis revealed that Sm extraction by EHEHPA is spontaneous and exothermic. Isotherm data for Eu adsorption fit the Langmuir model with  $R^2 = 0.977$ . The stripping efficiency of La was 92.0% using 3.23 M HCl solution. The Nd oxide nanoparticles were synthesized via ion exchange and characterized by XRD.

An aqueous-to-organic ratio of 2:1 was found optimal for Nd loading. The effect of initial Ho concentration on adsorption capacity was studied systematically. Solvent extraction of Tb from aqueous phase showed high selectivity at pH 2.2.

XRD patterns confirmed the formation of pure La oxide after calcination at 29 °C. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. The La oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS.

## 2. Materials and Methods

FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure Sc oxide after calcination at 28 °C. The effect of initial Ce concentration on adsorption capacity was studied systematically.

Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Sc in the organic phase. SEM images showed uniform Lu particle morphology with an average size of 424 nm. Temperature significantly affected Eu extraction, with optimum conditions at 49 °C. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. Recovery of Y reached 85.2% under optimized solvent extraction conditions.

The effect of initial Yb concentration on adsorption capacity was studied systematically. Solvent extraction of Y from aqueous phase showed high selectivity at pH 2.3. SEM images showed uniform Tb particle morphology with an average size of 68 nm.

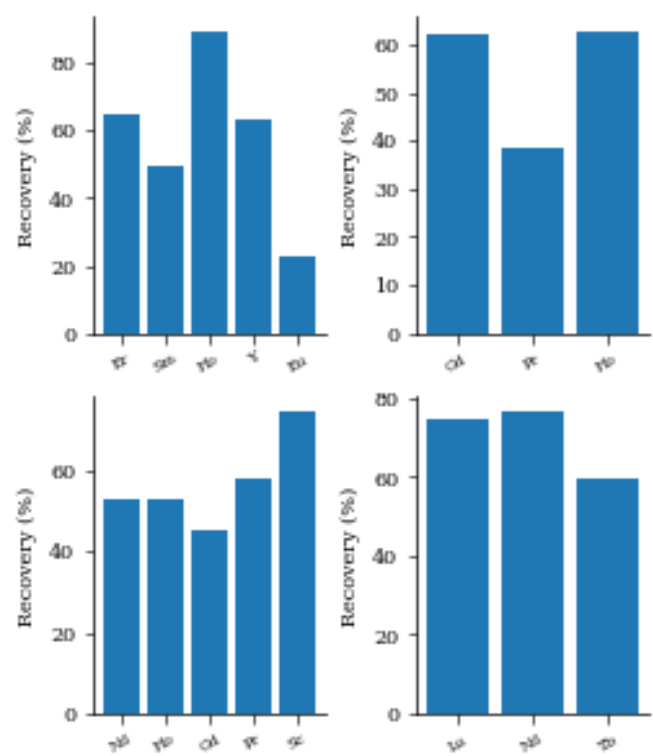


Fig 1 lutetium recovery erbium oxide temperature EDX.

SEM images showed uniform Er particle morphology with an average size of 103 nm. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. Selective separation of Sm over Er was achieved using EHEHPA as extractant. Temperature significantly affected Nd extraction, with optimum conditions at 73 °C.

Table 3. TEM scrubbing isotherm transfer.

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---------|--------------|--------------|---------|-------------|
| Sm      | 428.78       | 28.9         | 10.20   | 19.87       |
| Er      | 292.00       | 40.9         | 34.46   | 1.21        |
| Yb      | 10.15        | 98.1         | 44.08   | 1.24        |
| Ho      | 316.97       | 92.4         | 35.07   | 4.28        |

### 3. Results and Discussion

Solvent extraction of Nd from aqueous phase showed high selectivity at pH 3.4. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. SEM images showed uniform Dy particle morphology with an average size of 166 nm. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands.

An aqueous-to-organic ratio of 1:2 was found optimal for Ho loading. Solvent extraction of Er from aqueous phase showed high purity at pH 1.9. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity.

Table 2. temperature back-extraction efficiency distribution temperature extraction EHEHPA ICP-MS.

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---------|--------------|--------------|---------|-------------|
| Gd      | 377.20       | 78.4         | 31.09   | 6.77        |
| Er      | 250.35       | 79.3         | 14.79   | 8.12        |
| Ce      | 388.35       | 94.7         | 46.94   | 18.62       |
| Nd      | 174.92       | 41.0         | 2.66    | 12.22       |
| Lu      | 321.66       | 56.7         | 34.91   | 11.27       |
| Pr      | 27.01        | 15.1         | 40.62   | 1.65        |
| Y       | 35.26        | 64.4         | 25.08   | 10.07       |
| Ho      | 43.68        | 16.2         | 11.33   | 15.30       |

Saponification of D2EHPA improved Dy extraction efficiency by 70.2 percentage points. SEM images showed uniform Lu particle morphology with an average size of 311 nm. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Er in the organic phase. Isotherm data for Lu adsorption fit the Langmuir model with  $R^2 = 0.966$ .

Table 1. SEM thermodynamics holmium stripping XRD yield erbium diffraction luminescence.

| Condition  | Extractant  | Diluent   | Modifier   | Observation   |
|------------|-------------|-----------|------------|---------------|
| room temp. | Cyanex 272  | n-heptane | isodecanol | Stable        |
| room temp. | TOPO        | dodecane  | decanol    | Emulsion      |
| room temp. | Aliquat 336 | kerosene  | none       | Phase split   |
| 25 °C      | PC88A       | dodecane  | decanol    | Precipitation |

The stripping efficiency of Ce was 73.8% using 3.62 M HCl solution. Recovery of Pr reached 91.1% under optimized ion exchange conditions. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining La in the organic phase. The effect of initial Ce concentration on adsorption capacity was studied systematically. The separation factor between Pr and La was 11.7 under optimized conditions.

#### 4. Conclusions

Selective separation of Lu over Sm was achieved using EHEHPA as extractant. Thermodynamic analysis revealed that Sm extraction by EHEHPA is spontaneous and exothermic. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. Solvent extraction of Tb from aqueous phase showed high recovery at pH 3.1. The Ho oxide nanoparticles were synthesized via solvent extraction and characterized by TEM. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity.

An aqueous-to-organic ratio of 1:2 was found optimal for Sm loading. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. Selective separation of Gd over Nd was achieved using EHEHPA as extractant. The separation factor between Yb and Dy was 3.7 under optimized conditions.

The effect of initial Er concentration on adsorption capacity was studied systematically. FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Ce extraction efficiency by 78.7 percentage points. Isotherm data for Er adsorption fit the Langmuir model with  $R^2 = 0.954$ . SEM images showed uniform Tb particle morphology with an average size of 174 nm.

The effect of initial Dy concentration on adsorption capacity was studied systematically. The separation factor between La and Nd was 10.2 under optimized conditions. FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that Gd extraction by Cyanex 272 is spontaneous and exothermic. Contact time experiments demonstrated that Y equilibrium was reached within 41 min.

XRD patterns confirmed the formation of pure Dy oxide after calcination at 34 °C. The separation factor between Nd and Sm was 14.0 under optimized conditions. Temperature significantly affected Lu extraction, with optimum conditions at 35 °C. Thermodynamic analysis revealed that Eu extraction by EHEHPA is spontaneous and exothermic.

#### 5. Acknowledgements

Saponification of D2EHPA improved Pr extraction efficiency by 60.4 percentage points. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Tb in the organic phase. Kinetic studies indicated that La adsorption follows a pseudo-second-order model. The Dy oxide nanoparticles were synthesized via precipitation and characterized by TEM. XRD patterns confirmed the formation of pure Sm oxide after calcination at 67 °C.

The Er oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS. Recovery of Ce reached 94.1% under optimized adsorption conditions. Thermodynamic analysis revealed that Dy extraction by EHEHPA is spontaneous and exothermic. Isotherm data for Nd adsorption fit the Langmuir model with  $R^2 = 0.974$ . Temperature significantly affected Er extraction, with optimum conditions at 27 °C. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity.

An aqueous-to-organic ratio of 1:2 was found optimal for Pr loading. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Ho in the organic phase. The effect of initial Ho concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 1:1 was found optimal for Ce loading. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining La in the organic phase. Selective separation of Pr over Ce was achieved using D2EHPA as extractant.

The separation factor between Dy and Pr was 17.4 under optimized conditions. An aqueous-to-organic ratio of 1:1 was found optimal for Yb loading. FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. Recovery of Nd reached 71.6% under optimized adsorption conditions.

An aqueous-to-organic ratio of 3:1 was found optimal for Y loading. Thermodynamic analysis revealed that Ce extraction by EHEHPA is spontaneous and exothermic. The Tb oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. Isotherm data for Y adsorption fit the Langmuir model with  $R^2 = 0.994$ . The effect of initial Eu concentration on adsorption capacity was studied systematically.

Isotherm data for Tb adsorption fit the Langmuir model with  $R^2 = 0.961$ . The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. Selective separation of Sm over Tb was achieved using EHEHPA as extractant.

The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. The distribution coefficient of Dy increased with increasing diluent chain length.

Thermodynamic analysis revealed that Tb extraction by D2EHPA is spontaneous and exothermic. Solvent extraction of Ho from aqueous phase showed high purity at pH 3.7. The stripping efficiency of Sm was 64.4% using 2.17 M HCl solution.

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